

Test Project

Module A Mini Speed Test
Projects

Web Technologies

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Introduction

In this mini speed Test Project, you are given multiple mini tasks. We have collected all the tasks from multiple clients. And for priority reason, we have selected some tasks for this round of development. Please only implement the required tasks.

Please follow the instruction to know which one to implement. For each mini Test Project, there are three levels.

- Level 1: 0.5 points worth that is expected takes less than 15 minutes of work.
- Level 2: 1.0 points worth that is expected takes less than 30 minutes of work.
- Level 3: 1.5 points worth that is expected takes around 30 minutes of work.

Please put your files in /module-a/ and with the task ID as folder name, e.g.

/module-a/A1/ The XX is the workstation number.

Description of project and tasks

Please create an index page to link to each mini speed Test Project.

The index page should contain thumbnail and title of each mini speed test project.

Please write at least one git commit for each mini speed Test Project. The git commit messages should be readable and easy to understand.

Tasks for Design Implementation

A1: SCC Grid (Level 1)

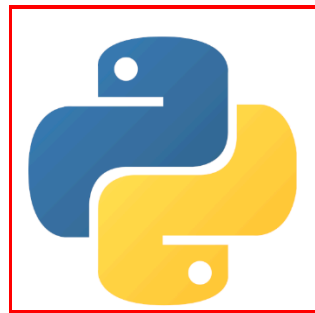
Implement the grid using CSS grid. Observing colors, fonts and internal indentation is not required.



A2: Vector image of logo.png (Level 1)

You have been provided with a bitmap image of logo.png. Create a vector version of it. Save the resulting file in one of the formats ai, svg, eps. The resulting file should not contain raster elements.

Example of logo.png (you can use 2 colors only):



A2 : Flag Animation :

Make the flag moving using html and CSS



A3: Preparing a web page for printing (Level 1)

You have been provided with a formatted web page.

You will need to change the provided html and css code so that when preparing for printing (for example, when we press Ctrl+P):

- all information should be in black and white (grayscale is allowed),
- there is only one color area (fragment/element) on the page - the logo,
- the font type is changed (for example, to Arial).

A4: A flower shape (Level 2)

Create a CSS-based flower shape consisting of 10 petals.



A5: A flower shape (Level 1)

Using the materials provided, create a banner in a graphic design software. The banner should be saved as a11.jpg with dimensions of 300x250 px. Ensure all materials are utilized and the text is legible.

Tasks for Front-end Development

B1: Image Compare (Level 2)

Given the HTML:

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <title>Image Compare</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div>
    <input type="text">
    <img src="">
    <img src="">
  </div>
</body>
</html>
```

You need to use the provided index.html, style.css (empty file), two images (car1.jpg and car2.jpg) and a splitter.svg file to implement the Image Compare functionality. The Image Compare

container has a splitter element. This element displays car1.jpg on the left and car2.jpg on the right. You can adjust the display ratio of the two images by clicking on the container or dragging the splitter element. As you adjust the ratio, one image will be displayed more prominently, while the other will be displayed less.

You cannot add or remove any HTML elements (except pseudo-elements). You can modify the attributes of existing HTML elements (including event listener attributes).

A sample video, "two-cars.mp4", is provided for reference. Below are some frames from the video:

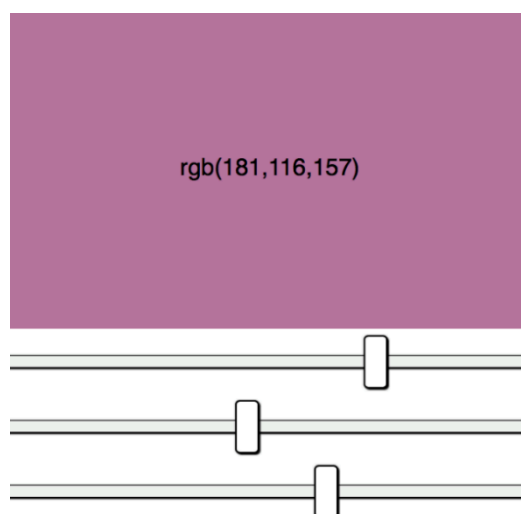


Requirement Notes

- The splitter left is car1.jpg, the splitter right is car2.jpg. No HTML elements were added or removed.
- The splitter can be moved by dragging with the mouse or clicking within the container.
- The image's effect is same as the two-cars.mp4 video.

B2: RGB slider (Level 1)

Implement the RGB slider as follows. There are 3 sliders to adjust the values of red, green and blue colors.



B3: Rounds (Level 2)

Create a page containing 5 circles. Each circle contains one of the following words: «Web», «PHP», «Javascript», «HTML», «CSS». The words are centered both horizontally and vertically.

The position of the circles on the page is arbitrary, but overlap is not allowed. The entire area of the circle should be visible. The page has an offset of 20 pixels from the edges of the screen. The size of the circles is also random, from 300 to 400 pixels. The position and size of the circles are random at each load.

The colors of the circles are: #1F2041, #4B3F72, #417B5A, #D0CEBA, #E9D2C0.

B4: Video Gallery Page (Level 3)

Create a Video Gallery page with provided videos.

The first video width is the full width of the page and the others are smaller than the first one like thumbnails.

All videos must have controllers with functions below

- Play
- Pause
- Full screen / Normal screen(When screen is full)
- A timeline to move to the desired moment of video
- Mute / Volume(When volume is muted)

You can use Fontawesome icons for the controller.

- Play : `<i class="fas fa-play"></i>`
- Pause : `<i class="fas fa-pause"></i>`
- Full screen : `<i class="fas fa-expand-arrows-alt"></i>`
- Normal screen : `<i class="fas fa-compress-arrows-alt"></i>`
- Volume : `<i class="fas fa-volume-up"></i>`
- Mute : `<i class="fas fa-volume-mute"></i>`

You should create controllers using HTML, CSS, JavaScript (including jQuery and JS Frameworks), not use original controls which video element provide.

Video file name should be shown on the top-left of video screen.

You can create your own design to improve user interface.

Tasks for Back-end Development

C1: Resize and Grayscale Multiple Images (Level 2)

The "images" folder contains several color images, each with a width of 640 px and varying heights (640 x N).

Your task is to edit all these images by:

- o Converting them to grayscale
- o Reducing the width to 320 px while maintaining the aspect ratio

Result: Save all the modified images in the "images-modified" folder.

C2: A class inheritance structure (Level 1)

Create a class inheritance structure that demonstrates the concept of polymorphism and encapsulation.

Task Description:

1. Create a base class named "Shape" with the following properties and methods:
Properties: color (string), filled (boolean)
Methods:
 - Constructor to initialize color and filled properties
 - Getter and setter methods for color and filled properties
 - Method named "getArea" that returns 0
2. Create two subclasses named "Circle" and "Rectangle" that inherit from the "Shape" class:
Circle class:
Properties: radius (float)
Methods:
 - Constructor to initialize radius property
 - Method named "getArea" that calculates and returns the area of the circle ($\pi * \text{radius} * \text{radius}$)
Rectangle class:
Properties: width (float), height (float)
Methods:
 - Constructor to initialize width and height properties
 - Method named "getArea" that calculates and returns the area of the rectangle ($\text{width} * \text{height}$)
3. Create an additional subclass named "Square" that inherits from the "Rectangle" class:
Square class should have a single property named "side" (float) and should automatically set the width and height properties to the same value when instantiated.

Requirements:

The classes should demonstrate proper usage of inheritance, encapsulation, and polymorphism.

The "getArea" method in each subclass should return the correct area based on the shape's properties.

Create instances of each class and demonstrate the correct calculation of areas.

You are encouraged to include additional methods or properties to enhance the class structure.

Submission:

Submit the PHP or Node.js code that implements the above class structure along with a brief explanation of how you approached the task and any additional features you included.

C3: Solution of equation (Level 1)

You are provided with an html page that contains fields for entering two real numbers k and b. Write a script that will run on the server and that will solve an equation of the form $kx + b = 0$.

The solution of the equation should be displayed on the page

Equation

Enter the coefficients of the equation $kx + b = 0$:

Coefficient k:

Coefficient b:

Solve

Result is:

Instructions to the Competitor

Each mini Test Project folder should be separated and self-contained.

Please include only exactly the required mini test project. And do not include any not related folders or not related mini Test Projects.

Other

This project will be assessed by using Google Chrome web browser.

Marking Summary

	Sub-Criteria	Marks
1	Mini Test Project General	6
2	Design Implementation	4
3	Front-end Development	4
4	Back-end Development	4